NATS CLI Extended

Login options

Login with username and password

```
nats --user <user> --password <pwd> sub foo
```

Login with token

```
nats --token <token> sub foo
```

Login with nkey file - must not contain line breaks

```
nats --nkey <file> sub foo
```

Login with creds file - operator mode - generated with nsc or Synadia Control Plane

```
nats --creds <file> sub foo
```

Storing options in local context

Save connection options to context

```
nats context add LOCAL --server nats://127.0.0.1:4222 --user <user> --password <pwd>
```

Select context

```
nats context select LOCAL
```

Quick validate all contexts (connect servers)

```
nats context validate
```

LLM support

Set `LLMFORMAT=1` in environment to switch to LLM friendly markdown output.

```
export LLMFORMAT=1
```

API Json Schemas

List all supported schemas used by client APIs to communicate with the server. This may include APIs not (yet) exposed in the standard clients.

```
nats schema search
```

Show the schema for the new ***multi get*** API

```
nats schema info io.nats.jetstream.api.v1.stream_msg_get_request
```

List and lookup server errors

List all API error codes emitted by the server

```
nats errors ls
```

Lookup details. The returned `Go Index Constant` is useful for tracing the error in the server code. LLMs in reasoning mode do a good job at identifying code paths leading to the error.

```
nats errors lookup <code>
```

Server Configuration

Obtain current server configuration and statistics (SYS)

```
nats server request variables
```

Reload server config file (SYS) - server id can be obtained with the command above

```
nats server config reload <server id>
```

Server and Stream Events

Connect/Disconnect and authentication errors

```
nats events
```

Including all Jetstream advisories and events

```
nats events --all
```

Graphs - ASCII Dashboard

A number of commands have a `graph` subcommand or `--graph` option.

```
nats server graph
nats stream graph <stream>
nats consumer graph <stream> <consumer>
```

Graph subject traffic. E.g. Jetstream consumer management.

```
nats sub '$JS.API.CONSUMER.CREATE.>' '$JS.API.CONSUMER.INFO.>' '$JS.API.CONSUMER.DELETE.>' --graph
```

Extended Health Check

Health probe (SYS) - used by Kubernetes et. al.

```
nats server report health
```

- Ensure you see all servers and all report 200
- Repeat multiple times if needed

Check clusters and routes (SYS)

```
nats server ls <server count>
```

- Ensure all servers respond quickly. This means all expected servers replied.
- Ensure there are 0 slow consumers
- Ensure routes/GW are consistent
- Look for consistent versions
- Look for consistent CPU usage
- Look for consistent uptime
- Look for consistent memory usage. Is one server using RAM excessively?
- Check the number of connections.

Message traffic (SYS) - Run as system user for extended info

```
nats traffic
```

- **Raft:** Ensure votes are all to 0 (non zero may indicate instability)
- **Raft:** Ensure append is half of reply
- **General:** Ensure there are less than 1000 JS API Requests per second

Jetstream health - Note that not all servers may have Jetstream enabled (SYS)

```
nats server report jetstream <JS server count>
```

- Ensure all servers respond quickly. This means all expected servers are there.
- Ensure the number of consumers is as expected
- Make sure there is a stable meta leader
- Compare the number of servers in the RAFT Meta Group with Jetstream summary. Ensure all server are current and show in the Jetstream summary.

Gather and analyze information for auditing

```
nats audit gather
```

- Keep audit data for post mortem.

And do a quick analysis

```
nats audit analyze
```

- This is not an exhaustive report. Details may change over time. See `nats audit checks` for a list of current checks.

Exhaustive Stream and consumer status (SYS) WARNING - These reports can be large and impact server performance

```
nats server stream-check
nats server consumer-check
```

- Check for out of sync streams and consumers. Use `--unsynced` to filter.
- Check for expected replicas count, stream size, message count and offline peers

The output of `nats audit gather` can be fed into `stream-check` and `consumer-check` for offline analysis

```
cat ./clusters/*/*/jetstream_info/* | jq -c '.' > jsz.json
nats server stream-check --stdin --unsynced < jsz.json
```

Jetstream queueing/bottlenecks (SYS)

Detecting bottlenecks in Jetstream replications and request routing

```
nats server request ipqueue
```

- Check **Routed JS API Requests** for pending API requests.
- Check RAFT groups for pending operations. Reports account, streams and consumer.

Detailed RAFT status report (SYS)

```
nats server request raftz --account='$G'
```

Identify suspect streams/consumers from `Cluster Group` in `nats stream info`

```
nats server request raftz --account=<account> --group=<raft group>
```

NATS CLI Extended

Configuring accounts and users in operator mode

The nats cli can now replace the `nsc` tool for most use cases.

- Cleaner and more intuitive syntax.
- More interactive modes for a quick start.
- Uses the same underlying library - *https://github.com/nats-io/jwt*
- Uses the same storage format on disk.

For details see:

- *https://docs.nats.io/running-a-nats-service/configuration/securing_nats/auth_intro/jwt*
- *https://docs.nats.io/running-a-nats-service/configuration/sys_accounts/sys_accounts*

Basic steps for setting up decentralized authentication. All configuration changes are stored locally until **`nats auth account push`**ed to a nats cluster.
Set **`XDG_DATA_HOME`** to override the local storage dirctory.

Create a new operator with system account
```
nats auth operator add SYSOP
```
Set SYSOP as default
```
nats auth operator select SYSOP
```
Generate a template server configuration file from an operator
```
nats server generate server.conf
```
Create an application account
```
nats auth account add MyAccount
--defaults
```
Create a new user
```
nats auth user add MyUser
--defaults
```
Create an admin user in system account
```
nats auth user add admin SYSTEM
--defaults
```
Export credentials for the system user
```
nats auth user credential
sys_admin.creds admin SYSTEM
```
Use **`nats context`** to set defaults
```
nats context add sysadmin
--description=SYSOP
--server=nats://localhost:4222
--creds=sys_admin.creds
```
Push an account or its changes from a specific operator to a specific server, using system account credentials
```
nats auth account push MyAccount
```

Monitoring with the CLI

nats server check provide simple warning rules, suitable to log file based monitoring. For proper monitoring the **NATS Prometheus exporter** or the **NATS Surveyor** should be used in conjunction with Grafana or similar tools.
List server check options and available warning rules. Checks one server at a time.
```
nats server check
nats server check server --help
```
Simple alerts on server CPU and memory usage (SYS)
```
nats server check server --name
<server> --cpu-warn=50
--cpu-critical=80
--mem-warn=100000000
--mem-critical=200000000
```
Alert on connection count being out of range. The check can be inverted by setting **critical** < **warn** (SYS)
```
nats server check server --name
<server> --conn-warn=1000
--conn-critical=2000
nats server check server --name
<server> --conn-warn=400
--conn-critical=300
```
Report in prometheus format (or nagios, json, text)
```
nats server check server --name
<server> --cpu-warn=50
--cpu-critical=80
--format=prometheus
```
Watch servers (SYS) and jetstream
```
nats server watch servers
nats server watch jetstream
```

Account management

Current account information, connections overview and statistics
```
nats account info
nats account report connections
nats account report statistics
```
Backup and restore all streams in an account. See **`nats backup --help`** for stream backups, edit and validate.
```
nats backup account <directory>
nats backup restore account
<directory>
```
Purge/Delete all streams in an account. (SYS)
```
nats server account purge
<account>
```

Advanced Jetstream management

Change the cluster leader of a stream or consumer. Useful when stream or consumer are unresponsive
```
nats stream cluster step-down
<stream>
nats consumer cluster step-down
<stream> <consumer>
```
Change to a preferred leader
```
nats stream cluster step-down
<stream> --preferred <server>
```
Rebalance **all** stream leaders in the account - **Use with caution**
```
nats stream cluster balance
```
Scaling a stream down and up
```
nats stream edit <stream>
replicas=3
```
Catching a stream create config json with --json or --trace
```
nats stream add <stream>
--defaults --subject 'foo.>'
--json
nats stream add <stream>
--defaults --subject 'foo.>'
--trace
```
Creating a stream from json config
```
nats stream add --config <config
file>
```
Reset a consumer. Without **`--sequence`** it rewinds to the ack floor and redelivers all unacknowledged messages
```
nats consumer reset <stream>
<consumer>
```
Reset to a stream sequence, e.g. to skip a poison message. Consumers with a start sequence or start time cannot go below it.
```
nats consumer reset <stream>
<consumer> --sequence=<seq>
```

Advanced cluster management

Rebalance client connections between servers (SYS) - **Use with caution** and check the extended options with --help
```
nats server cluster balance
```
Change meta raft leader (SYS) for repairing a cluster
```
nats server cluster step-down
```
Removing a node from a cluster after a cluster scale down (SYS). NATS cluster will wait for nodes to come back. Streams will not repair their replicas until nodes are removed - **Use with caution**
```
nats server cluster peer-remove
<server>
```
If the server name is unknown find and use the server id (SYS)
```
nats server report jetstream
nats server cluster peer-remove
<server id>
```

Created by Synadia, the team behind NATS - www.synadia.com
https://github.com/roeschter/cheat-sheet-nats-synadia - 2026.11.09

NATS CLI Extended

Rescue a meta group that lost quorum (SYS, 2.15+) - e.g. 2 of 3 servers lost for good. **Use with caution**

Confirm there is no meta leader and which servers are missing

```
nats server report jetstream
```

Temporarily lower the quorum on the surviving servers. Prompts for the quorum size. Use `--js-domain` if servers run in a JetStream domain.

```
nats server cluster rescue
--js-domain <domain>
```

Within **5 minutes**, before the lowered quorum expires, remove the dead servers

```
nats server cluster peer-remove
<server> --js-domain <domain>
```

Quick Server status reports

Note: some reports are accessible from the SYS account only or show different details in SYS and app account

The server health probe as exposed by the HTTP interface and used by K8S et.al. to report server readiness.

```
nats server report health
```

Jetstream status, including meta raft group status.

```
nats server report jetstream
```

Quick memory and cpu usage report

```
nats server report cpu
nats server report mem
```

Detailed server status, profiling and connection kick

Most information is also available on the **http debug (port 8222)** interface. Most of this information is auto collected by `nats audit gather`. Except for **connections** all commands require a system account.

List all connections.

```
nats server request connections
```

List connections and all subscriptions

```
nats server request connections
--subscriptions
```

Kick a connection. (SYS)

```
nats server request kick <conn
id> <server id>
```

Current server configuration. (SYS)

```
nats server request variables
```

Detailed status report per server of all streams and replicas. (SYS)

```
nats server request jetstream
```

All subscriptions system wide (SYS)

```
nats server request
subscriptions --detail
```

.. other reports are **accounts**, **gateways**, **leafnodes** , **routes** and **raft**

Various GoLang runtime profiles. (SYS) - Supports: allocs, heap, goroutine, mutex, threadcreate, block, cpu

```
nats server request profile
<profile>
```